MCDB138 – MIDTERM 2 REVIEW

DISCUSSION 6

A Few Points

- Midterm 2 – Tuesday, May 11
  - *Lectures 2 – 10 (mostly 6 – 10)*
- More free response questions
  - *Less questions total*
  - *1 hour*
- Extra OH – Friday (11am – 12pm)
- Paper 4 discussion (next week)

- Questions during the review will be based on teams! ☺
  - *Try your best to participate! It helps with remembering terms!*
  - *Also, feel free to pass your question to another team!* 😉

Categories

- Pattern formation (A/P and D/V axis)
- A/P axis and zygotic genes
- Cell movement and rearrangement
  - *Cytoskeleton*
  - *Gastrulation*

Pattern Formation

- What are maternal effect genes? Give some examples
- Genes present in the embryo during oogenesis and before fertilization examples are Bicoid, Caudal
- They are responsible for the expression of zygotic genes.

Most of the maternal effect genes are morphogens because they act in a concentration dependent manner to specify cell fate.

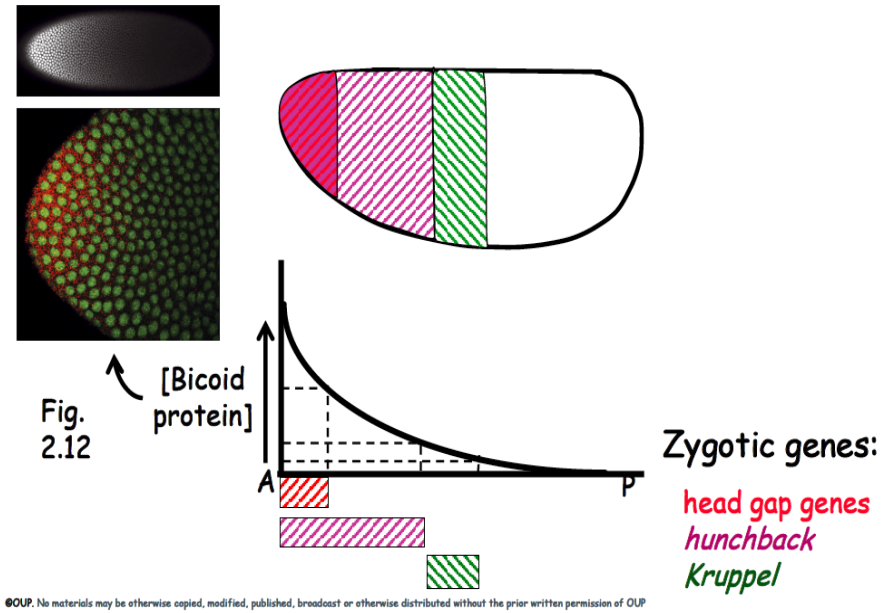


Fig. 2.12

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- How is maternal mRNA localized in the cell during oogenesis?

- Microtubules and motor proteins
  - *Kinesins and dyneins are motor proteins that couple ATP hydrolysis to walk along the microtubules.*

- So, – end is anterior and + end is posterior *(relating to how motor protein walk)*

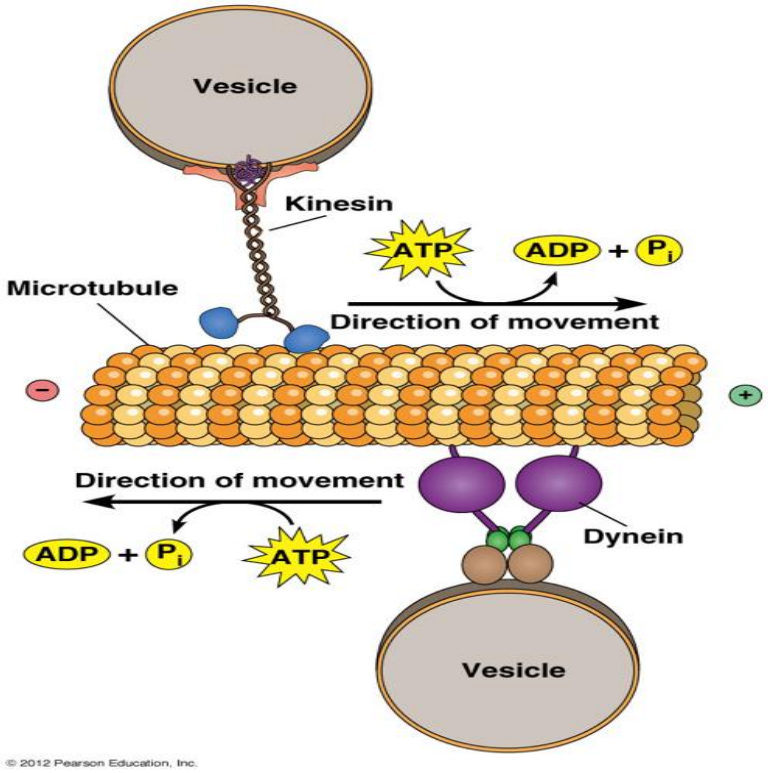
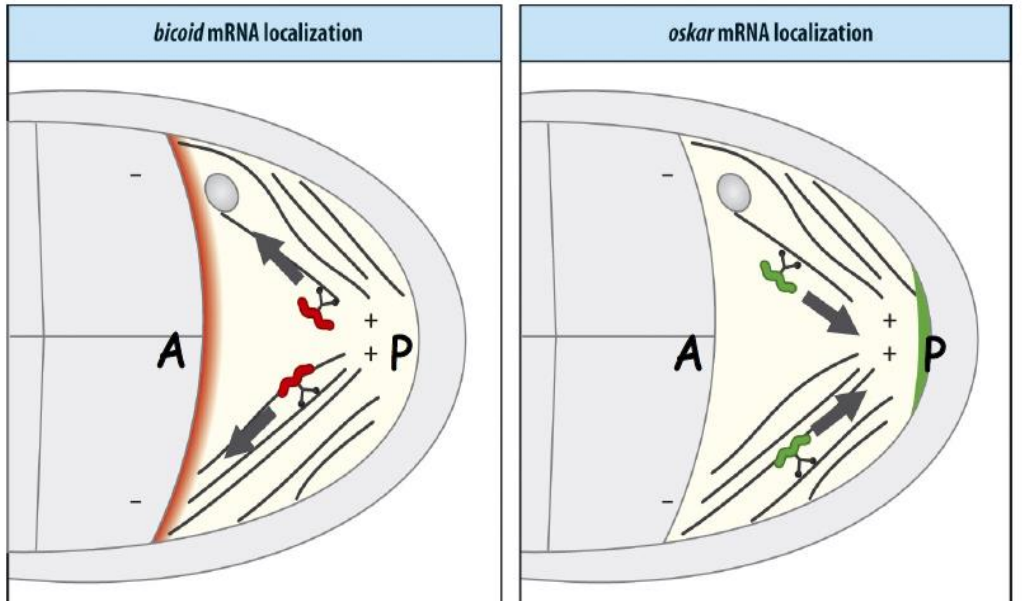


Fig. 2.20

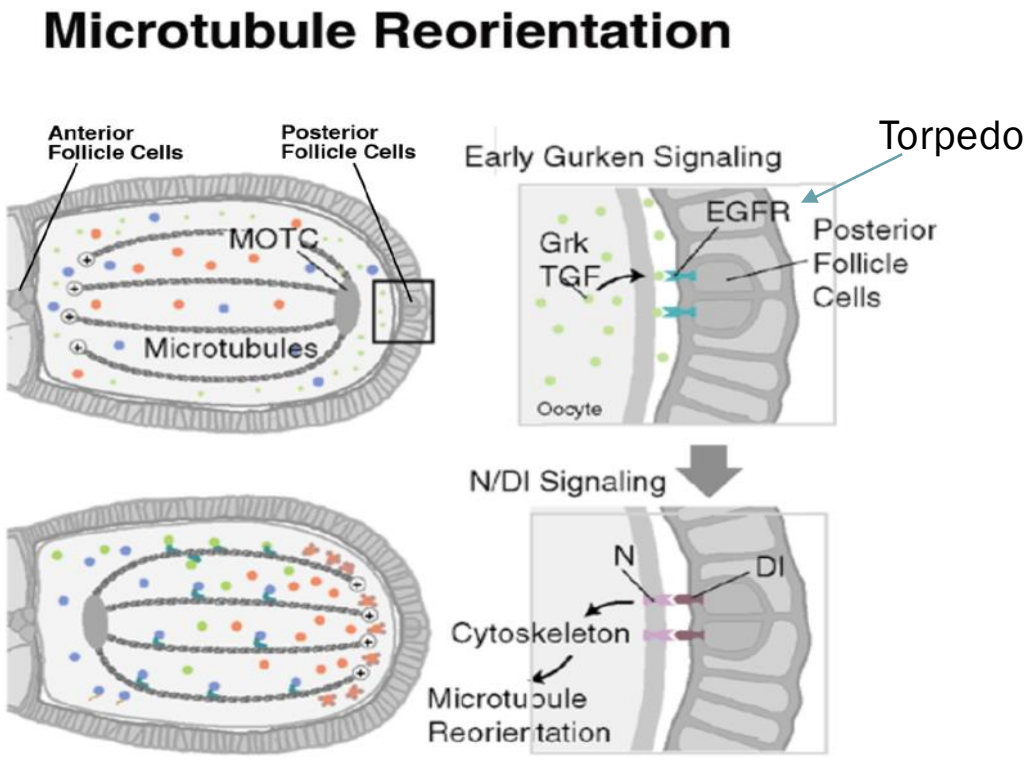


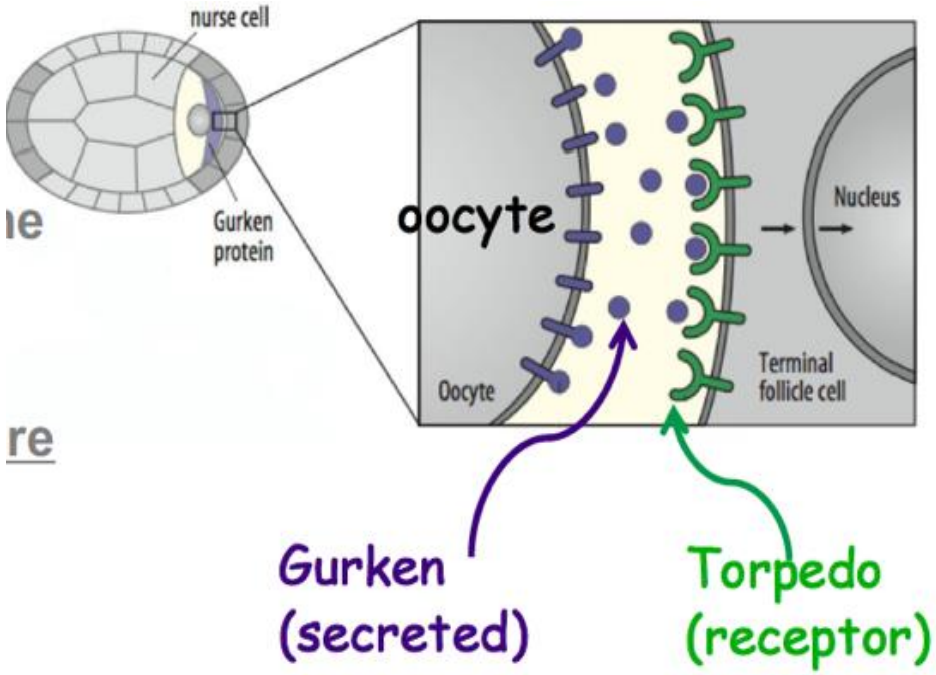
- What is responsible for the early specification of the potential A/P axis before Bicoid localization?

- Gurken signaling that leads to microtubule reorganization

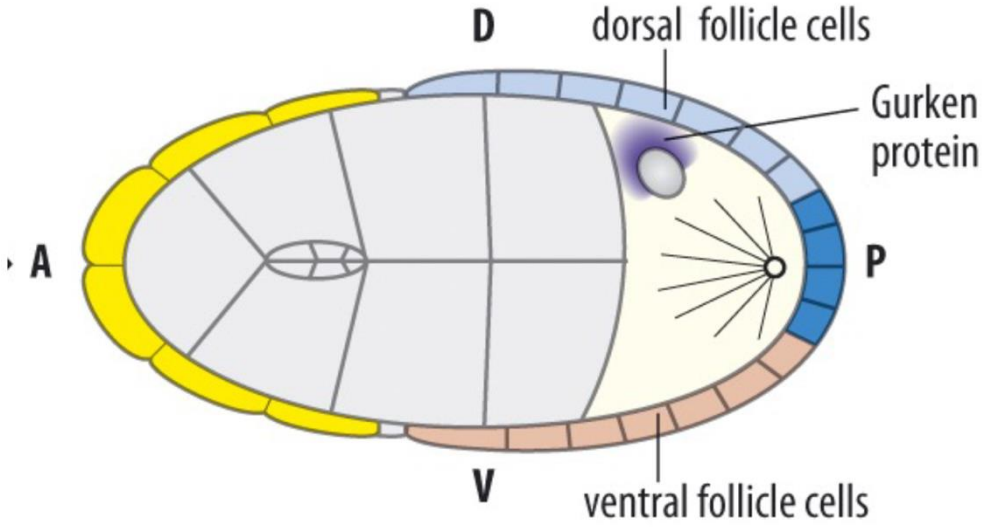
- Note: Gurken signaling is known to occur in two phases based on what it does
  - *Gurken signaling for Microtubule reorientation (AP axis)*
  - *Gurken signaling for Dorsalization of the follicle cells (DV axis)*

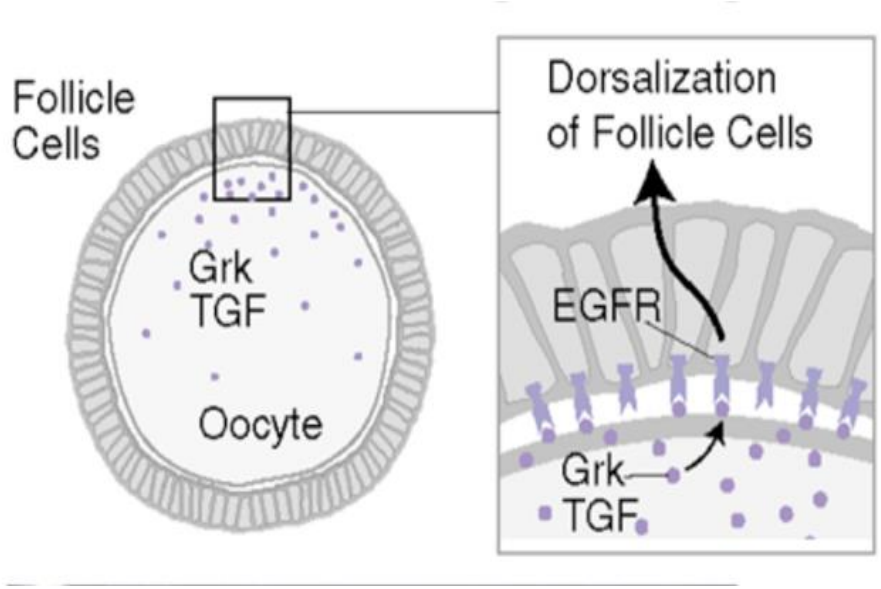
- Gurken binds to torpedo on adjacent follicle cells. Torpedo is an EGFR



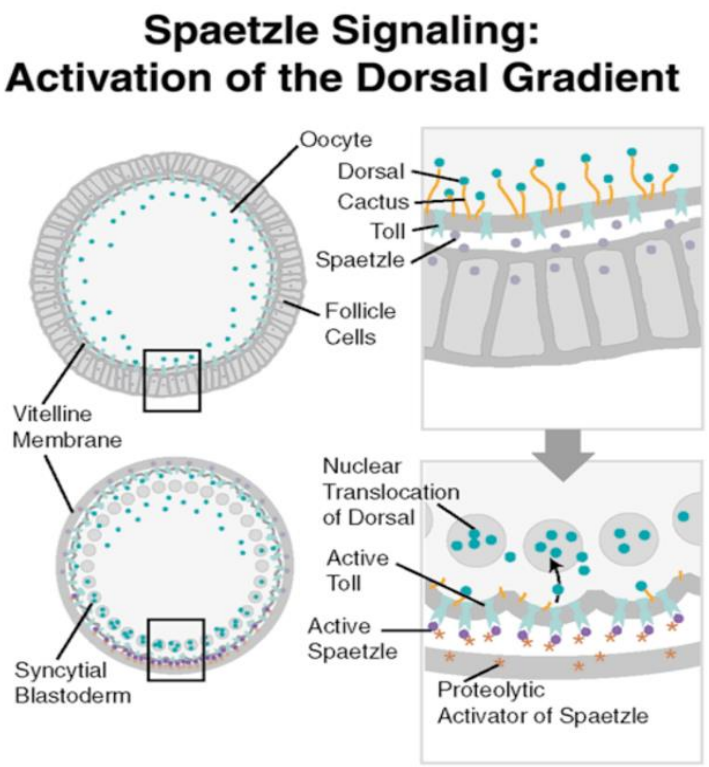


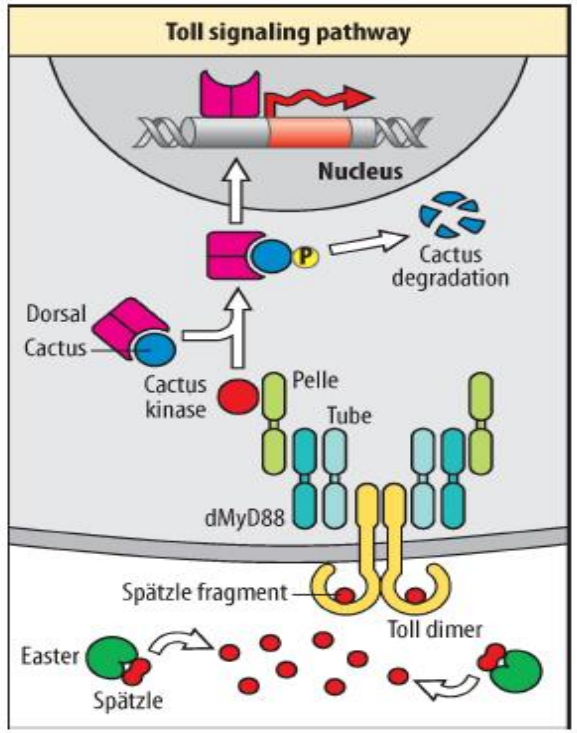
- The MT reorganization leads to the translocation of the nucleus to the future dorsal side
- Gurken mRNA deposited by nurse cells moves to the region around the nucleus
- Gurken binds to adjacent follicle cells and specifies them as dorsal





- What happens after Gurken signaling?
- Since D/V axis is somewhat specified;
- Ventral follicle cells only produce spaetzle, a ligand for toll receptor which is actually uniformly distributed in the egg but only activated by spaetzle
- Toll binding to spaetzle leads to phosphorylation of cactus which allows Dorsal protein enter the nucleus

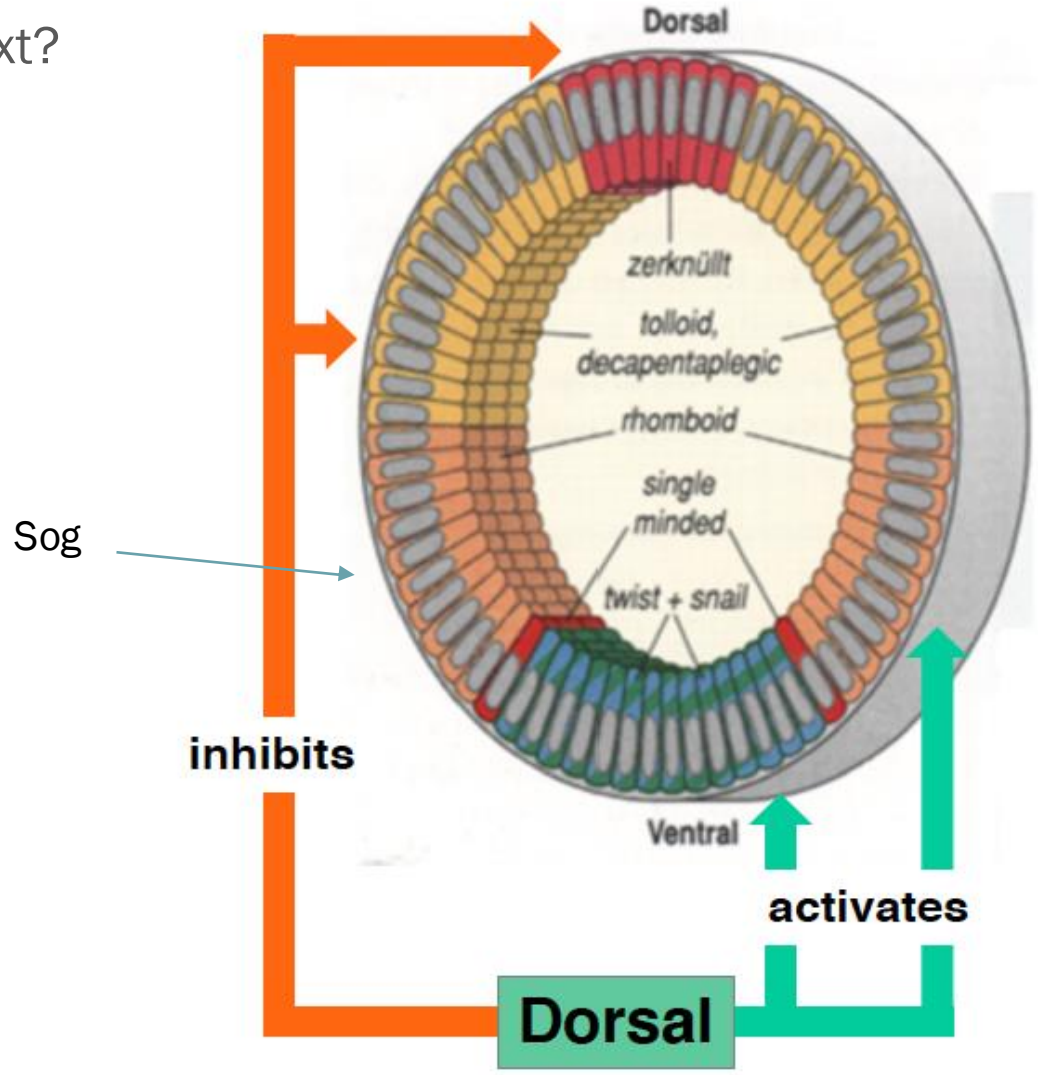




Box 2B

- After dorsal is activated, what happens next?
  - *Twist*
  - *Dpp*
  - *Tolloid*
  - *Sog*

- Map showing what dorsal protein controls



Takeaways!

- Understand what specifies the different axes (A/P, D/V)
- Understand the genes that are expressed in the axes
- Understand what will happen if the protein levels are increased or decreased
- Understand what will happen if these genes are knocked out.

A/P axis and zygotic genes

- Important to note!
  - *The genes and what controls what*
- Follow the arrows
  - *Added an arrow from maternal to pair-rule genes. Next slides will give more context!*

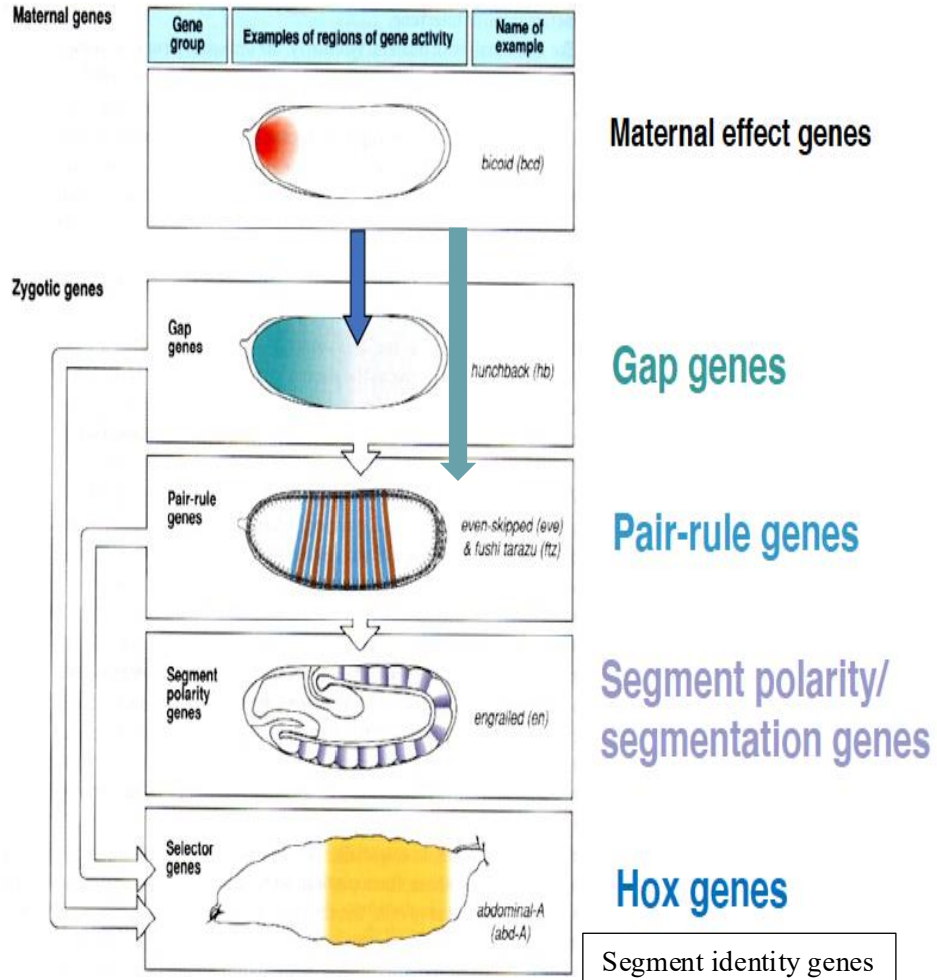


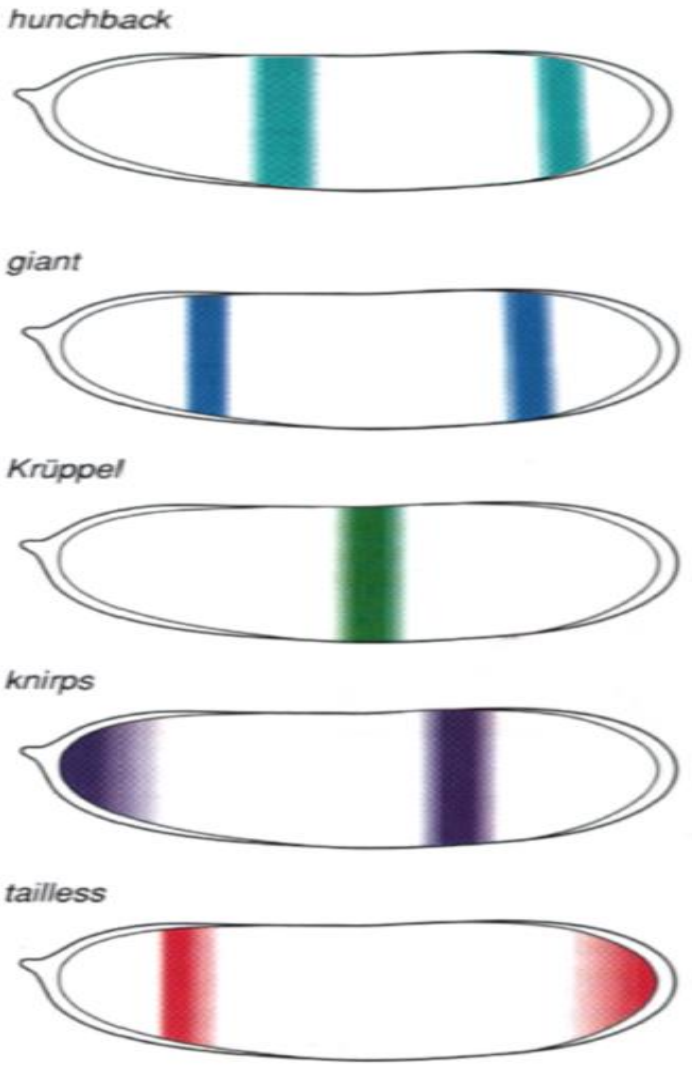
Fig. 2.9

Gap genes

- Gap genes are controlled by?
- Bicoid and also other gap genes. They have a cross-regulatory mechanism that control their borders (repression)

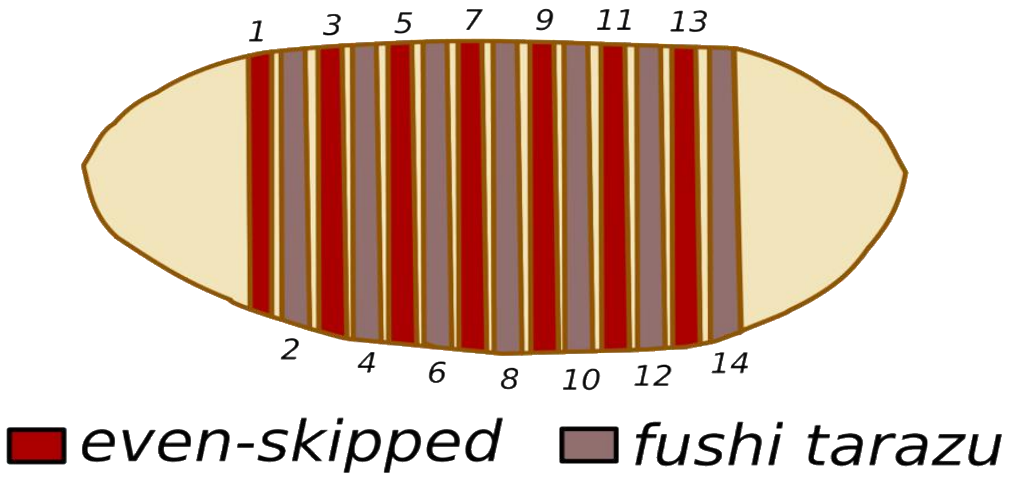
- Interesting fact:
- They are named gap genes because their loss will lead to large gaps in the embryo
- Which answers the question about which will affect the embryo more, loss of gap or pair-rule genes

- So, what will happen if Kruppel is KO. What about tailless?



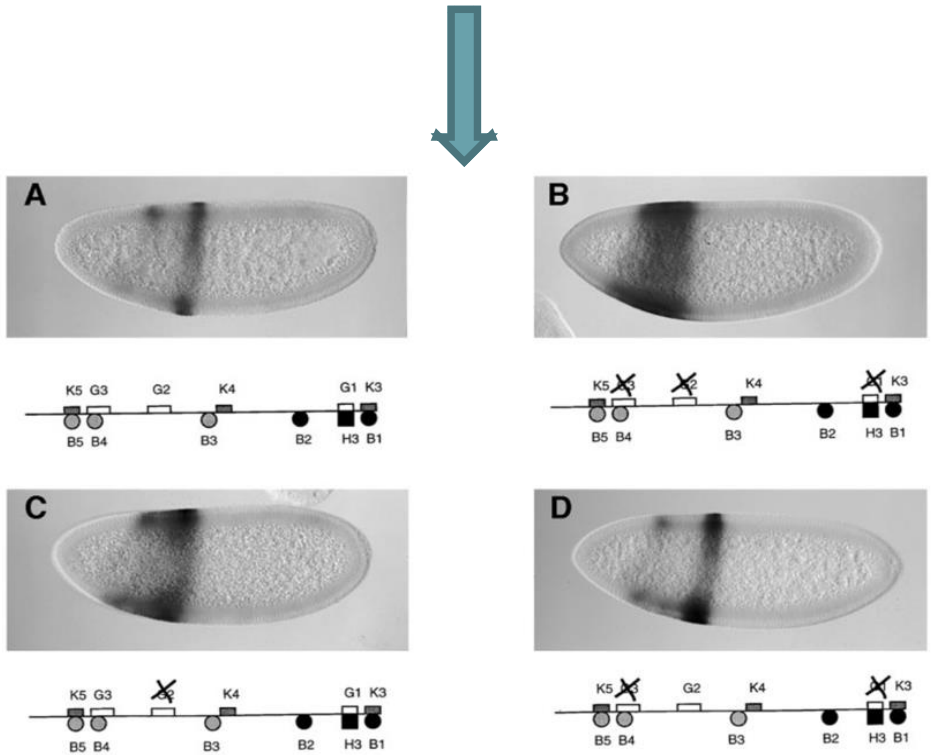
Pair-rule genes

- Pair-rule genes also have a cross regulatory mechanism to specify boundaries (repression)
- Also named based on the mutation

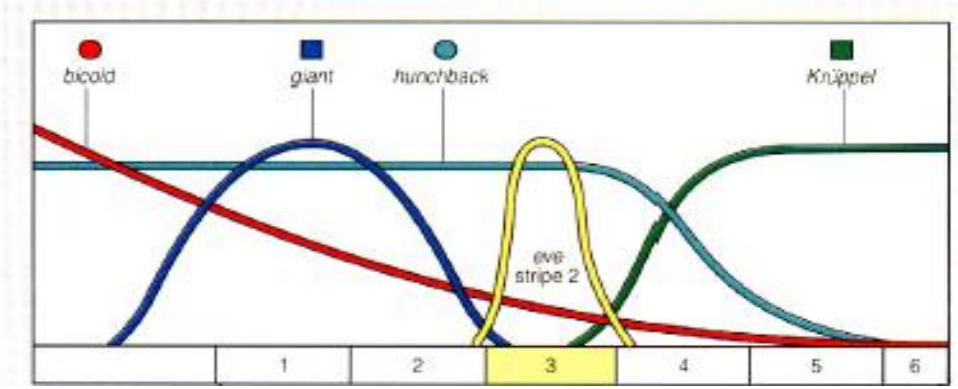


- What is the molecular mechanism that controls gap and pair-rule gene expression?
  - *Enhancers e.g., stripe 2 enhancer*
  - *Expression of eve in stripe 2 is controlled by a small piece of regulatory DNA called the stripe 2 enhancer.*

- Study showing how the maternal and gap genes regulate pair-rule genes with eve stripe 2
- Gt and Kr (gap genes) repress expression of eve while Bcd and hb are activators. For the stripe 2 enhancer!



B- Bicoid (5 Bicoid sites)
H- Hunchback
K- Kruppel
G- giant



This domain corresponds to eve stripe 2. Other pair-rule genes and gap genes are controlled in a similar manner by other enhancer modules in their regulatory domain.

Segment polarity genes

- Which segment polarity gene establishes the anterior boundary? Posterior?

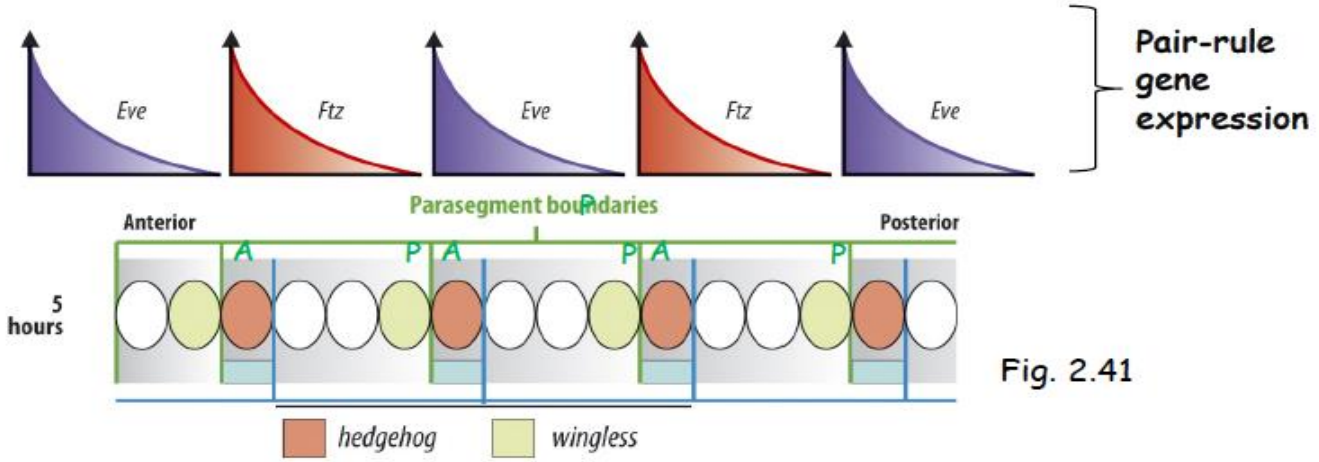
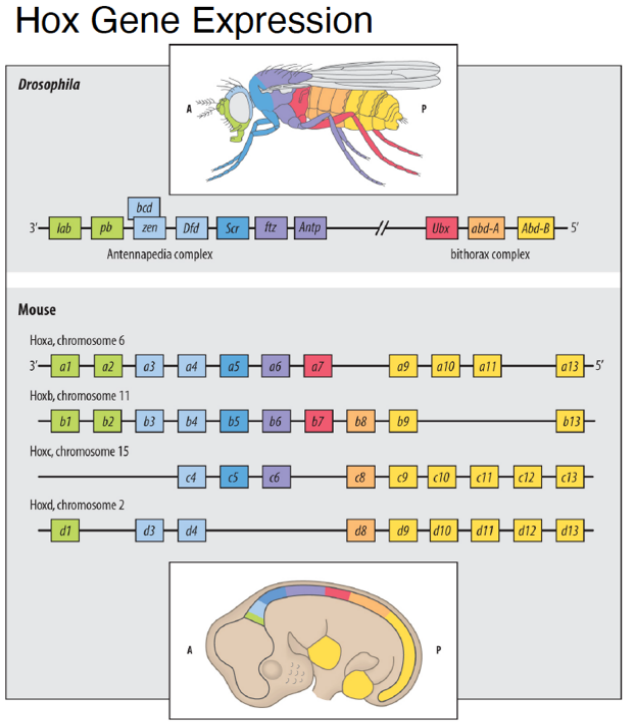


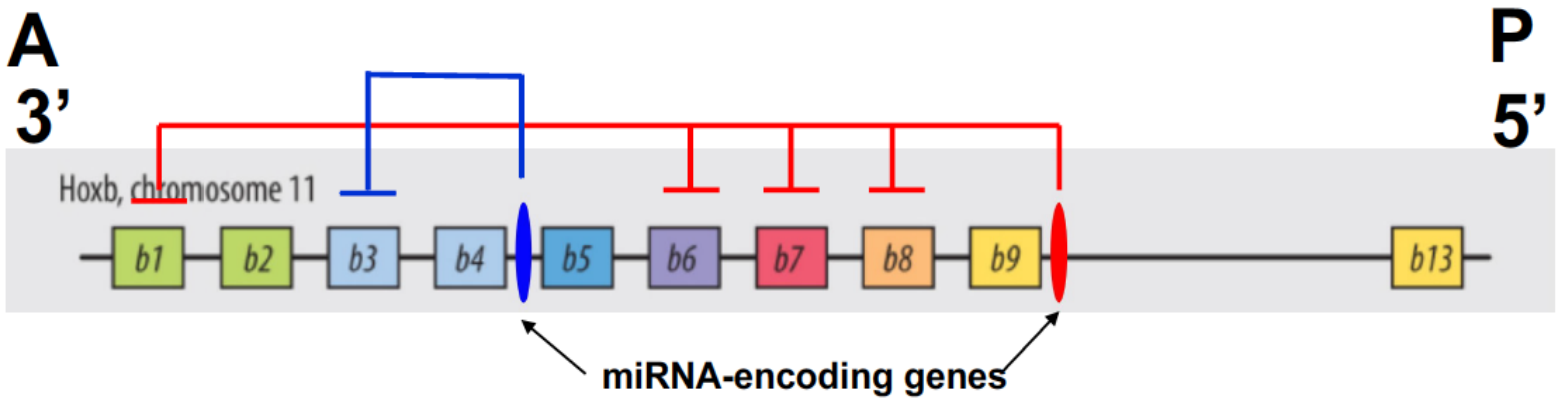
Fig. 2.41

Segment identity genes

- Important take away points;
  - *Expressed from 3’ to 5’. That means anterior Hox genes are expressed earlier*
  - *If a posterior hox gene is deleted, more anterior hox genes will be expressed and vice versa*
  - *Spatial and temporal expression (order and time)*
  - *Conserved- order, function*
  - *RA – can induce activating chromatin modifications*

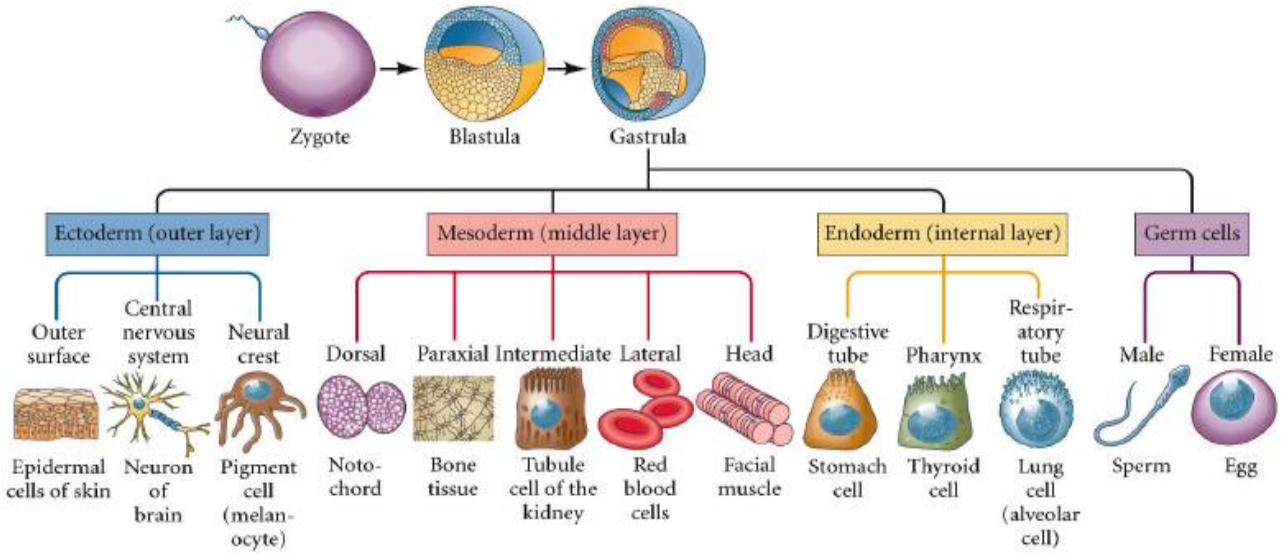


- Asides the gap genes and pair-rule genes, how is collinearity of Hox genes maintained?
  - *Chromatin modifications (Box 8B of the textbook is really helpful for understanding chromatin modifications)*
  - *miRNAs*



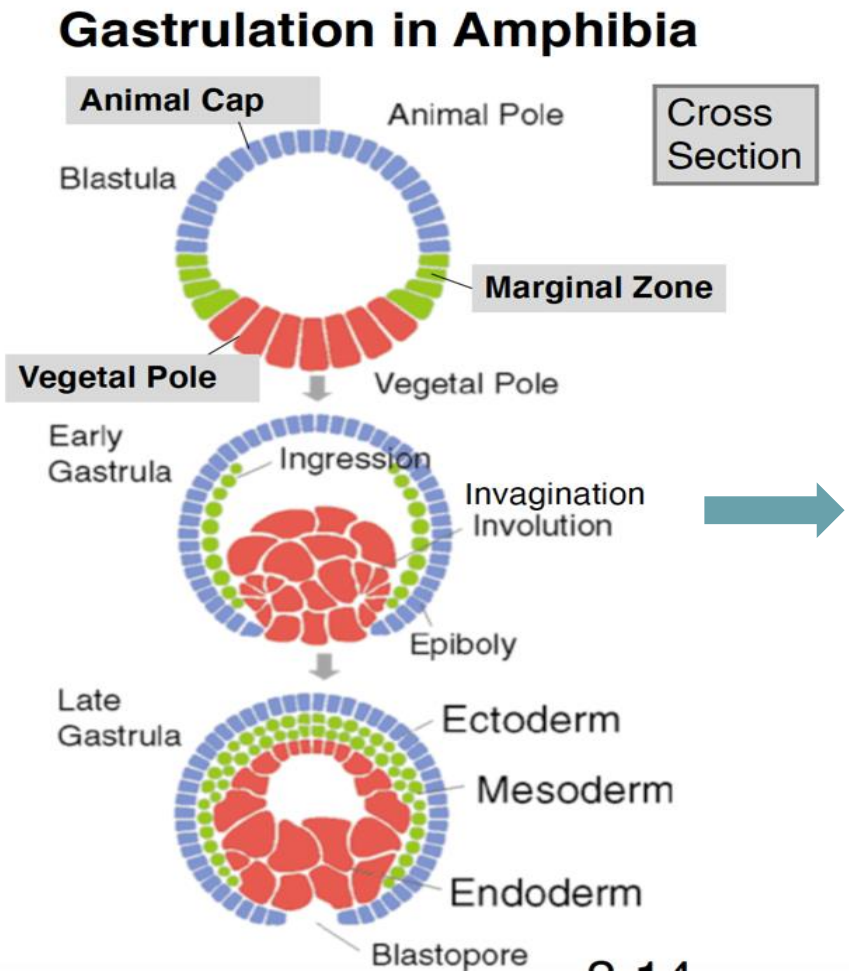
Cell movement and rearrangement

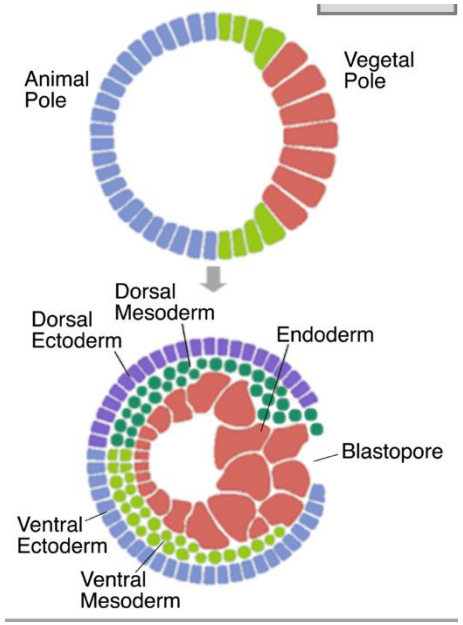
- Gastrulation- formation of the 3 germ layers – endoderm, mesoderm, ectoderm

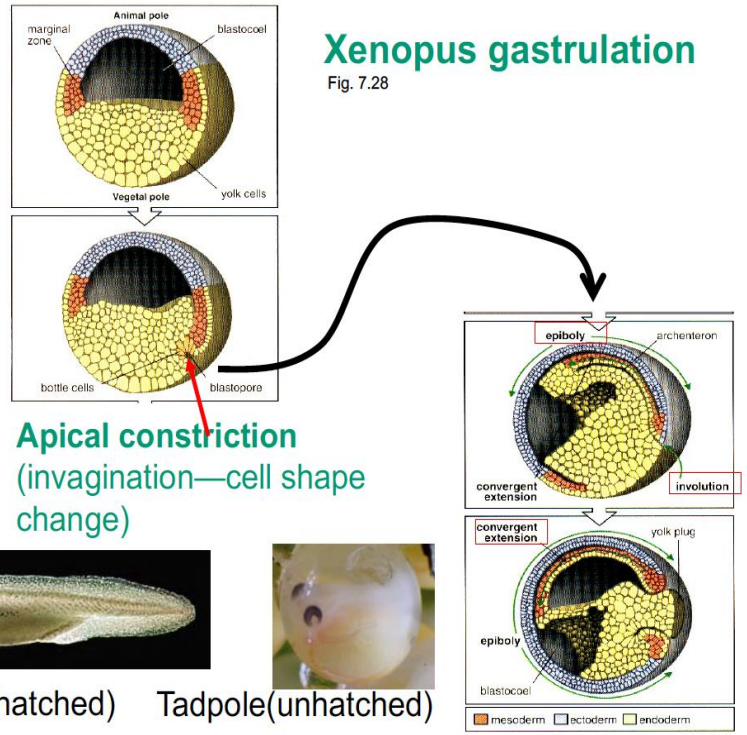


- What are the cell movements responsible for gastrulation?
  - *Invagination*
  - *Ingression*
  - *Epiboly*
  - *Involution*
  - *Convergent extension*

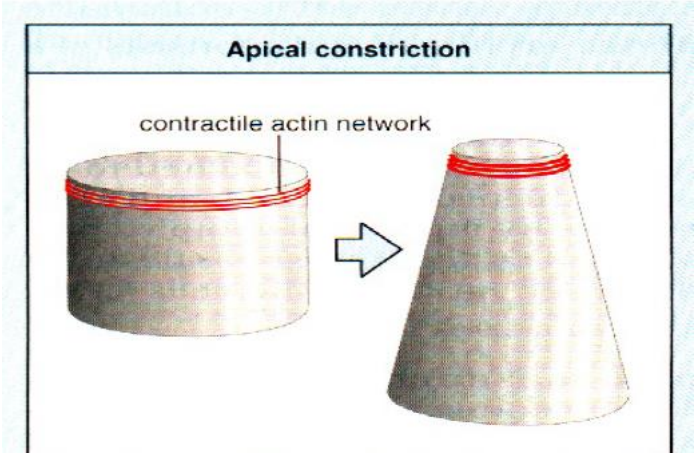
An example to help us visualize the movements



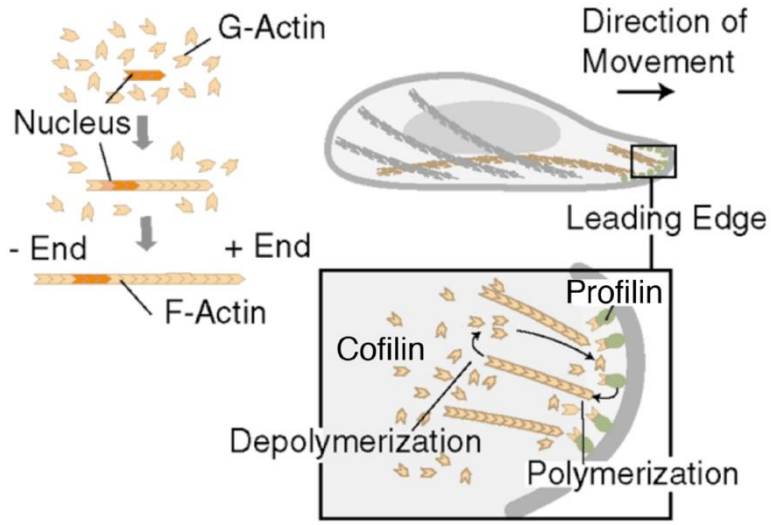




- How are these cellular movements made possible?
  - *Cytoskeleton*
- Actin – myosin
  - *Contraction*
  - *Cell movement*
  - *Cell shape*

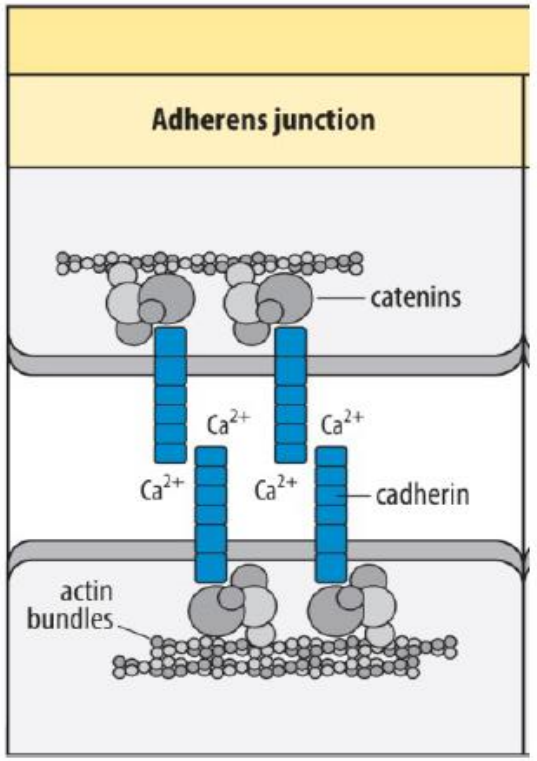


- How does actin control cell movement?
  - *Polymerization and depolymerization of actin filaments (treadmilling)*

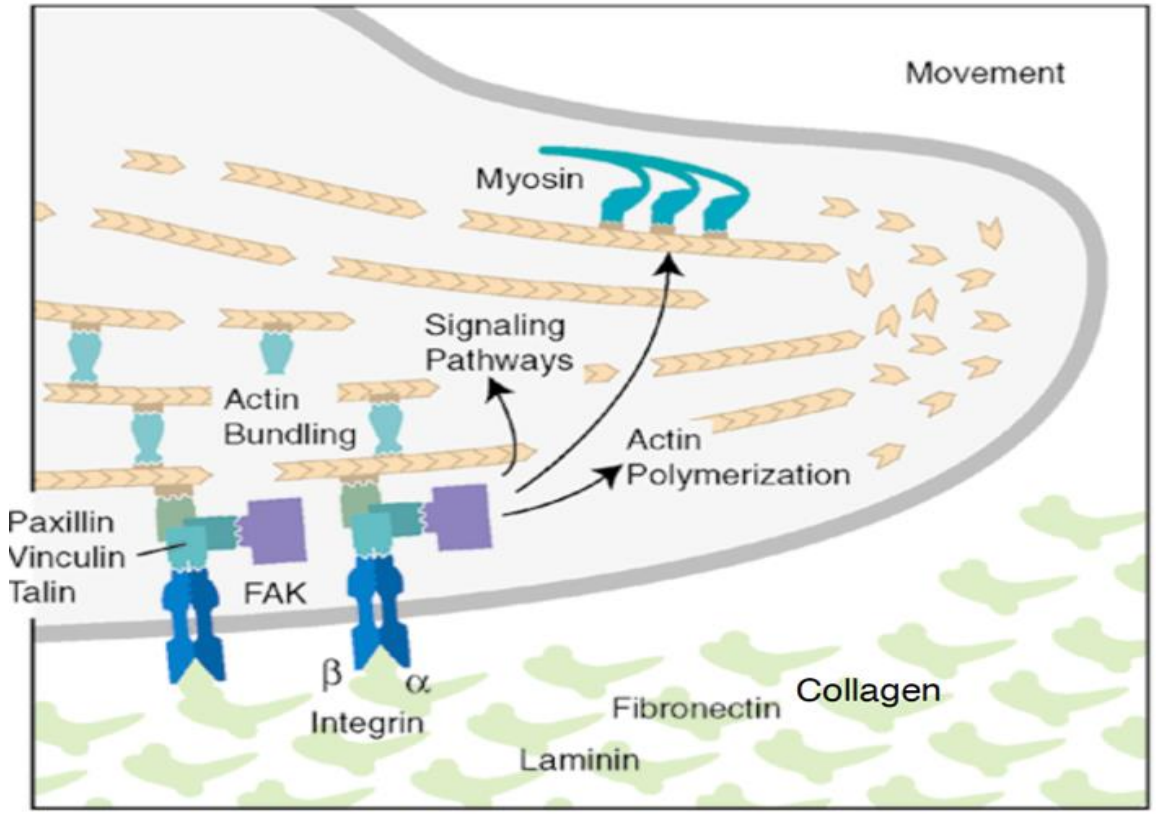


- What differentiates epithelial cells from the mesenchymal cells?
  - *Cell-cell adhesion*

- Cell- cell interactions
  - *Adherens junctions (cadherins)- homophilic*
  - *Tight junctions (occludins)*
  - *Gap junctions (connexins)*



- Cell- ECM interaction
  - *Integrins (heterophilic)*
    - Aids cell movement



Final Question! 🥳

- What happens during EMT and MET?

